

Choose one of the six extended essay options and upload a one- or two-page response. Please include the prompt at the top of the page.

- 1. What if the moon were made of cheese? Or Neptune made of soap? Pick a celestial object, reimagine its material composition, and explore the implications. Feel free to explore the realms of physics, philosophy, fantasy...the sky is the limit!**

—Inspired by Tate Flicker, Class of 2025

Mom handed me a shrink-wrapped square brick. Rounded corners, Minnie Mouse's face shining through the thin layer of plastic. It felt a bit squishy. "Open it! Here, you have to put it in water," she exclaimed, leading me and my twin brother to the kitchen sink. Peeling away the plastic, I felt the fibrous, soft almost, texture of the slab. I dropped it into the water, and, quite anticlimactically, it started expanding, unfolding into a small, wrinkled, drenched towel. What was I supposed to do with a wet towel? Still, I put on a facade of wonder. I can't lie, it was definitely... unimaginable. My childish excitement when seeing one of these just a few months ago at a thrift store almost makes me wish I'd been more enamored with the "magic towel" as a kid. It's a weird phenomenon. To me, nostalgia is always thrilling, no matter the mundanity of the recalled moment.

Maybe it's the sound of the name, the human inclination to root for the underdog, or its endearing biographical description that comes up on google, "Our favorite dwarf planet since 2006," but I've always been fond of Pluto. First learning about its demotion to "dwarf" made those normal days we learned about it as the ninth planet in our solar system in Montessori school seem so suddenly important, nostalgic.

For someone with limited astronomical knowledge, I've thought about Pluto quite a bit in my life. Sure, it was the respected and prestigious International Astronomical Union that ruled Pluto a "dwarf planet," and sure, it was astrophysics and planetary scientist Neil Degraffe Tyson that stated it would be an insult to other planets to deem Pluto a planet, but what if there was something they didn't know? Say, Pluto was made of viscose rayon, the fiber of regenerated cellulose of which the magical towels of my childhood were made of.

It doesn't rain on Pluto, so it just hasn't had the chance to expand. Scientists have only speculations about Pluto's chemical and material composition: a rocky core with a mantle of ice, and a buried ocean beneath the ice surface. You may retaliate against my theory by pointing out that if Pluto is composed of ice or has an ocean, water has already come in contact and Pluto has failed to expand. But, the ice on Pluto has been compared to the rock that makes mountains on our Earth. It's Pluto's "bedrock;" it doesn't move or melt. Furthermore, Pluto's ice is formed by gaseous methane frost that condenses into ice directly on Pluto's surface. This layer of ice, pressurized by Pluto's atmosphere, is what could be presumably preventing a magic towel core from enlarging.

I still think Pluto has a chance to get promoted, to expand to its full form and take its rightful place as ninth in our planetary system. What we call fireballs are actually extremely bright and burning meteors. If enough of these meteors come in contact with Pluto's surface, its methane ice surface would evaporate into gas depressurizing and exposing its buried ocean, allowing water to seep into the viscose rayon fiber. If nickel-sized magic towels with 0.750 inch diameters can expand into towels of 9 inches, then Pluto's core, approximately 70% of its diameter, could expand from 1700 km to 20,400 km. Pluto would become the 5th largest planet

in our solar system, become almost twice the size of our beloved Earth, still maintaining around a 527,000 km distance away from planet #8, Neptune. While towel material lacks the rigidity you'd expect a planet to have, Pluto's atmosphere would continue to gradually turn methane frost into ice, solidifying Pluto's expanded form. That seems pretty deserving of the "planet" title to me.

I imagine that the day of Pluto's promotion will be a nostalgic one for the world: #DwarfNoMore trending on Twitter, the satisfaction of being able to use the original nine-word acronym I learned to remember the order of the planets, and Mr. Tyson begrudgingly denying his status as a "Pluto-hater."

How does the University of Chicago, as you know it now, satisfy your desire for a particular kind of learning, community, and future? Please address with some specificity your own wishes and how they relate to UChicago. (no word limit)

Transferring from a small public school deep in the San Fernando Valley to a K-8th school for gifted students overlooking LA was like switching worlds. A personal MacBook awaited me inside my new Apple-sponsored 4th-grade classroom. We had typing tests instead of spelling quizzes. Two students behind me celebrated their 75 wpm. Scoring a zero myself, I aspired to become comfortable with this foreign 13-inch piece of hardware that was now mine. Opening up my laptop at home, I was mesmerized by the ability to color-code desktop folders and "inspect element" on webpages. I started with Lego Robotics and Scratch, but the programming classes I took in middle school inspired me to push myself. I branched out to new languages: XCode and Unity at Hackathons, HTML, CSS, and Javascript, which I learned on the job as a Product Design Intern at an AI company, GumGum, and implemented at my junior year hackathon.

Computer science soon transcended my programming classes. In ninth grade biology, I learned about computational biology and the intersections between machine learning, gene editing, and bioinformatics. As we talked about the gene-editing technology, CRISPR, we also delved into the ethical questions behind designer babies. I became enchanted by the ethical debates and started to think about the possibility of a dangerous supercomputer, the ethical duties of artificial intelligence researchers, and joined our high school's Ethics Bowl Team to engage in exciting discussions and learn to logically present my thoughts. Not only did I see computer science in aspects of all my classes, but I also tried to incorporate my own interests into CS. For my last Hackathon, I decided I wanted to learn how to use Spotify's API to create customizable playlists based on someone's mood. Just the idea of creating a music-based program made the 36 hours of nonstop programming bearable and exciting because I had hoped that the combination of two of my passions would result in a rewarding product.

As a student at the University of Chicago, I'd be able to continue building my computer science skills while applying and exploring my interests in sustainability, music, engineering, ethics, and the natural sciences. With classes like Introduction to Computer Security, Mobile Computing, Applied Artificial Intelligence for Materials Science and Engineering, and Engineering Interactive Electronics onto Printed Circuit Boards, I could both improve my

computer science skills and learn how to apply them to practical and tangible subjects. I'm also excited by the Core Curriculum's Philosophical Perspectives I course, which would allow me to take a deep dive into the world of ethics and philosophy, setting me up to more effectively engage in computer science classes like Ethics, Fairness, Responsibility, and Privacy in Data Science. I also deeply appreciate UChicago's approach to interdisciplinary learning. Classes like The Mathematics of Evolution give me the perfect opportunity to learn about the intersection between mathematics, the environment as it relates to ecology, evolution, genetics, and computer science. Nowhere else have I seen a better representation of my interests or a possible path to pursue all the subjects I am passionate about studying.

As someone who prioritizes sustainability in my daily life, I appreciate UChicago's commitment to sustainability within the community. I value UChicago's available resources like the Recycling Directory and initiatives like UChicago ECo and EF Campus.

At UChicago, I'd get to keep pursuing music as a student through courses like Electronic Music I, which attracts me since I want to make music independently. Not only do I have the option to take music electives, but UChicago's Core Curriculum makes music and the arts an integral part of students' education with core courses like Music Performance as Laboratory. Outside of taking advantage of the university's courses, I want to participate in The Performance Program and collaborate with others who share my passion for vocal music through acappella groups like The Ransom Notes. Being part of my school's Jazz Singers, a group singing mainly acappella music, I have a deep appreciation for performing live music.